

# Pedro Henrique Amorim Rezende

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## Professional Summary

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Backend and Platform Engineer with experience building and operating cloud-native distributed systems in logistics and telecommunications environments. Skilled in Python, Kotlin, Kubernetes, Kafka and microservices architectures, with hands-on experience working with scalable platforms and high-performance backend services. PhD in Computer Science with a strong foundation in distributed systems, networking, and software engineering. Proven track record working in international environments and collaborating with global technology partners across large-scale infrastructure and product-driven organizations.

## Core Skills

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|---------------------------|-----------------------------------------------------------------------------|
| <b>Languages</b>          | Python, Java, Kotlin, C/C++, Javascript, Shell Script                       |
| <b>Frameworks/Library</b> | Django, Micronaut, React                                                    |
| <b>Architecture</b>       | Microservices, Event-Driven Systems, Async Processing                       |
| <b>Protocols / APIs:</b>  | REST APIs, gRPC                                                             |
| <b>Cloud / Infra</b>      | Kubernetes, Kafka, Terraform, AWS, VirtualBox, VMware ESXi, Linux, CircleCI |
| <b>Databases</b>          | PostgreSQL, MongoDB, Elasticsearch                                          |
| <b>Observability</b>      | Prometheus, Grafana, Kibana, Jaeger, Wireshark                              |

## Professional Experience

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### Singapore Institute of Technology (SIT)

*Research Fellow / System Engineer*

Singapore

Jan 2023 – Jul 2025

- Managed traditional and Open RAN 5G infrastructures supporting industry and academic research across cybersecurity, robotics, drones, and video analytics, enabling the successful deployment and validation of 20+ real-world 5G use cases.
- Collaborated with partners such as HPE, Nokia, Viavi, Keysight, and Rohde & Schwarz in production-like operational environments.
- Led a benchmarking collaboration with Viavi to evaluate traditional and Open RAN infrastructures, defining test methodologies, troubleshooting system-level issues, executing performance assessments, and analyzing results. Outcomes were presented with Viavi at the Industrial Presentation Session of IEEE Globecom 2025, increasing industry visibility and engagement with the testbed ecosystem.
- Served as the main technical interface between SIT and external 5G system integrators, namely ST Engineering and HPE, coordinating deployment, troubleshooting, validation, and end-to-end testing activities.
- Led the integration and validation of partner solutions by configuring network elements, VPN servers, routing/switching policies, QoS parameters, and DNN configurations.
- Worked with distributed and cloud-native systems using Kubernetes, Open5GS, free5GC and OAI.
- Performed troubleshooting and observability activities using Wireshark, Jaeger, ROMES, Qualipoc, and Grafana to analyze control/data traffic and investigate connectivity and handover issues.
- Reviewed high- and low-level system designs and validated commissioning, functional, and end-to-end testing activities.

### Loggi Tecnologia

*Software Engineer*

Brazil

Jul 2020 – Dec 2022

- Developed and maintained backend APIs and services in Python and Kotlin within a microservices architecture supporting 300,000+ daily logistics operations across Brazil.
- Implemented synchronous and asynchronous integrations with databases, messaging brokers, and internal services supporting logistics operations at scale.

- Implemented performance optimization solutions using bulk upserts, indexing strategies, caching, query optimization, and read replicas.
- Enhanced observability and traceability by enriching events in an event-driven infrastructure and improving Kafka dead-letter queue filtering workflows.
- Contributed to design/code reviews, automated testing, deployment, and production support activities.
- Developed operational web interfaces using Javascript and React to streamline package handling workflows and improve operator productivity.
- Applied infrastructure changes using Terraform/Atlantis and supported Kubernetes-based environments.

## Additional Technical Experience

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**Visiting Researcher – University of Coimbra, Portugal** 2019 – 2020

- Designed and developed a solution using Java-based Floodlight and Python-based Mininet to reduce energy consumption in software-defined networks.

**Visiting Researcher – Concordia University, Canada** 2018

- Extended C-based Open vSwitch datapaths and Java-based SDN controller APIs to support multi-stream traffic splitting in OpenFlow networks.

**Intern – Federal University of Uberlândia** 2013

- Worked with PHP, shell scripting, Apache/Tomcat servers, SNMP, digital certificates, and authentication systems.

## Education

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**PhD in Computer Science** – State University of Campinas (UNICAMP), Brazil 2016 – 2022

**MSc in Computer Science** – Federal University of Uberlândia (UFU), Brazil 2014 – 2016

**BSc in Computer Science** – Federal University of Uberlândia (UFU), Brazil 2009 – 2013